
Positional Prediction: Consonant Cluster Prediction Text Entry Method for Burmese (Myanmar Language)

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Abstract

I am investigating a consonant cluster prediction text entry method for Myanmar language based on positional vowel information. The concept of this method is adapted by inspecting and carefully considering the nature of Myanmar language word formations or hand writing orders. According to the initial user study with our software keyboard prototype, users were able to type at ~41.57 characters per minute with mouse, ~41.84 characters per minute with stylus pen, ~14.76 characters per minute with Logitech dual thumbstick game pad and ~20.52 characters per minute with Nintendo Wii remote controller. These numbers are sure to improve as further refinements on program interface are made. The merits of the proposed predictive text input method is that even first time users can type Myanmar sentences with appropriate typing speed, and that this approach is applicable for various kinds of mobile devices (see Figure.1) and extendable for other similar syllabic languages such as Khmer (language of Cambodia), Nepali (language of Nepal), Thai (language of Thailand) and Hindi (one of the languages of India) etc.

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Figure 1. Four directional arrow keys on Nokia N76 mobile phone, Dell X51 PDA and Sony PSP portable game player

Keywords

Text entry, text input, predictive text entry, Myanmar language, Positional Prediction, syllabic languages

ACM Classification Keywords

H.5.2. Information interfaces and presentation: User interfaces – *Input devices and strategies*

Introduction

Although many alternative text entry methods exist for English, Japanese and Chinese etc., there is no efficient method for Myanmar language mobile devices. Currently, we only have PC keyboard based text input interfaces (e.g. onscreen QWERTY based keyboard) or a big screen keyboard that displays all of the Myanmar characters (i.e. consonants, dependent vowels and vowel signs, Independent vowels subscripted characters and abbreviations) as buttons [1]. QWERTY based keyboards are originally designed for English, and not directly usable for non Roman scripts such as Myanmar language (Burmese). And users have to spend time on searching each Myanmar character with big screen software keyboards and that will affect typing speed. Furthermore, it is also not appropriate for mobile devices that have limited screen and keypad. In this paper, I propose “consonant cluster prediction” text entry method based on given vowels positional information. “Positional Prediction” is a new typing concept for Myanmar language, however, the result of the user study proves that it is applicable and user-friendly. By providing meaningful and pronounceable consonant clusters during typing process, typing error rate can go down. To my knowledge, predicting consonant clusters with positional vowel information is the first proposal not only for Myanmar but also for other similar syllabic languages.

Myanmar Language

Myanmar script has been a major language of Myanmar for over 1000 years old, and it is the official language there. It belongs to the Tibeto-Burman language family and derives from Sino-Tibetan. Myanmar alphabets adapted the Mon script in the 8th century. Myanmar language has various types of characters comparing with English. i.e. consonants, dependent vowels or medials, independent vowels, finals, tones and subscript characters or conjunction alphabets etc. And Myanmar language contains many Pali (one of the Prakrits of ancient India and the language in which Gautama Buddha preached) words especially for religious things such as praying. Overall writing direction is from left to right, and the word order is SOV (Subject+Object+Verb). In a Myanmar sentence, spaces are used to mark phrases, not dividing words. An example of Myanmar word formation for the word “marriage” can be seen in Figure.2.

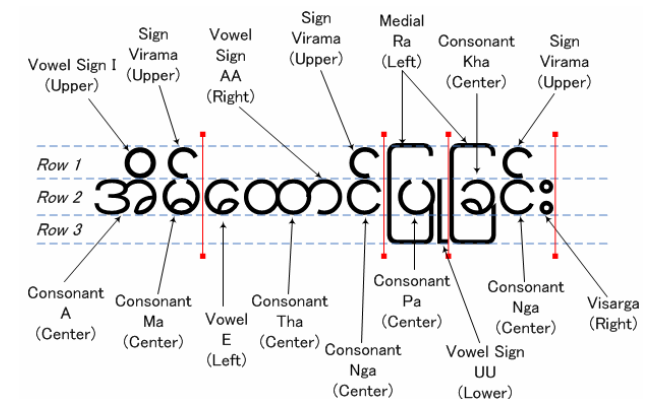


Figure 2. Word formation in Myanmar language (vertical lines indicate the pronunciation break)

Related Works

The research work for predictive text entry system began in China and Japan around 1960s and gradually got popular together with word processing software. Some predictive methods are based on phonetics (e.g. Romaji (or) Kana codes in Japanese and Pinyin codes in Chinese). Other methods are based on the shape of Chinese characters (called Kanji in Japanese) or stroke orders (e.g. Four Corner Method, Wubi and Q9 etc.)[2].

The dictionary based predictive methods like Tegic's T9 and Zi Corp's eZiText predict the word or phrases according to the key sequences type by users. For example, when user press "6" (mno) and "3" (def) on a standard ISO mobile phone keypad, the system will predict the words like "of" and "me", and then order by usage frequency. Ambiguity arises when more than one word matches a given key sequence (e.g. "5477" = "kiss", "lips") and "next" key is used for choosing other alternative words. Or users can stop predicting system and spell the word which involves rewriting the whole word using multi-tap. To improve the performance of disambiguation algorithms, some research work has been already proposed such as dictionary based predicting with used context of the phrase or document being typed [3] and LetterWise [4] that used prefix-based disambiguation instead of using dictionary of stored word by MacKenzie, Kober, Smith, Jones and Skepner (2001).

On the other hand, traditional multi-tap text input method is still popular because of its simple nature. As far as I know, there is no predictive text input that supports Myanmar language yet.

Concept of Positional Prediction

In making an analysis of Myanmar word formation, I found that writing system largely depends on adding left, right, upper and lower characters to consonant (i.e. consonant clusters). Here, left, right, upper and lower characters mean Myanmar dependent vowels, directives and subscript consonants that are always written with a consonant (e.g. left: "၉" (E) and "၉" (RA), Right: "—၁" (AA) and "း" (VISARGA) etc., Up: "၀" (I), "၀" (II) and "၀" (ANUSVARA) etc. and Down: "၁" (U), "၁" (UU), "၁" (HA THO), "၁" (HA THO TA CHAUNG NGIN), "၁" (WA SWE), "၁" (WA SWE HA THO), "၁" (AUKMYIT) and "၁" (Subscripted KA) etc.). They have to be written always with consonants (i.e. dependent), and their positions are defined when they are combined with consonants. Taking this into consideration, I propose a new consonant cluster prediction method based on given positional information of dependent characters. For example, "Ka + Right" for "က", "ကး", "ကျ", "ကျာ" and "ကျး", "Ka + Down" for "က၁", "က၁" and "က၁" and "Kha + Down + Up" for "က၁" etc. The vowel positional information adding order can be "Consonant + Left + Down + Up + Right" or Consonant + Left + Up + Down + Right. This text entry order is similar to Myanmar Unicode input order. Logically, it can also support hand writing order (i.e. Left + Consonant + Down + Up + Right or Left + Consonant + Up + Down + Right). Users only need to mention positional vowel information once even for the words that need to combine more than one positional vowel. Based on the given positional vowel information, the system calculates all possible consonant cluster combinations, then, removes impossible vowel combinations for given

consonant (i.e. unpronounceable or no meaningful combinations), and after that, sorts possible consonant clusters with average usage frequency. Here, note that there is no proper or standard usage frequency table for Myanmar characters and words yet and I use my own usage frequency table, which is the result of my previous work. Finally, the system sorts the consonant cluster according to users' typing history. The example process of positional prediction for "Kha + Left + Right + Up" can be seen in Figure.3.

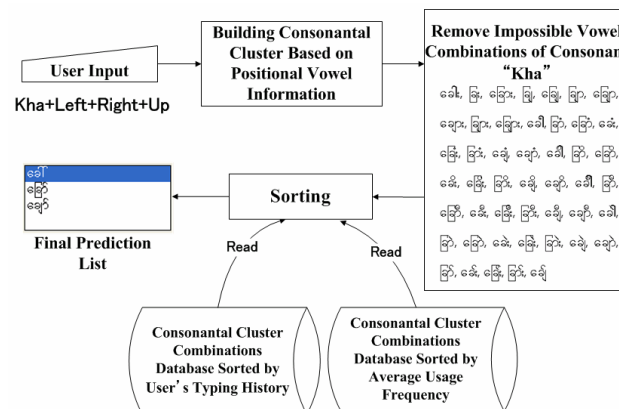


Figure 3. Process flow of Positional Prediction for Myanmar consonant “Kha” with vowel information (Left + Right + Up)

The Prototype

Here, I present Positional Prediction text entry prototype for Myanmar language, which was developed with Microsoft Visual Studio .Net 2003. The prototype can be seen in Figure.4. To reduce the candidate selection time, I just mention a self subscript character of given consonant in predictive list. Consonants subscripting with different characters and Myanmar

short form writing can be typed with subscript button
(i.e. the button labeled with " x_2 " character).

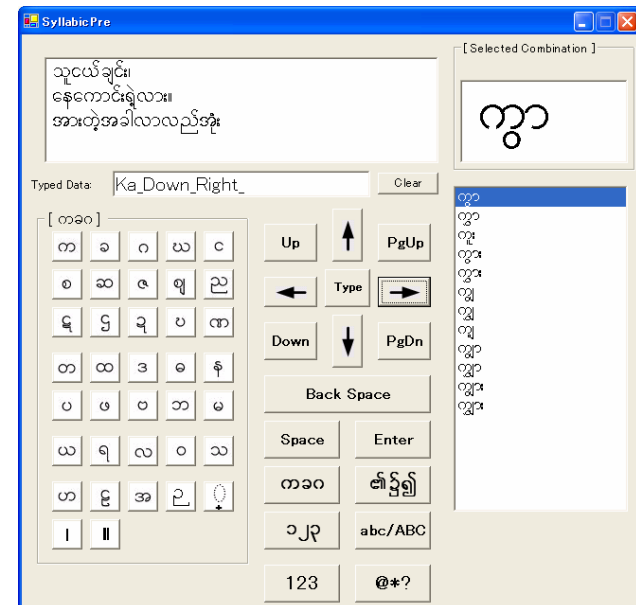


Figure 4. Positional Prediction text entry prototype for Myanmar language (Up, Down, PgUp and PgDn buttons are used for moving the highlight selection of consonant clusters of list box)

This prototype is usable with mouse, pen stylus, Logitech dual thumbstick game pad and Nintendo Wii remote controller. With dual thumbstick game pad, "left thumbstick" works for selecting consonants and other editing buttons and "right thumbstick" works for giving Left, Right, Up and Down information (see Figure.5).



Figure 5. Dual thumbstick game pad that we used for user study (left thumbstick is use for consonant and right thumbstick is use for positional vowel information inputs).



Figure 6. Wii remote controller (four directional keys are used for vowel position information)

With Wii remote controller, the mouse pointer can be controlled by moving the remote controller and the four directional keys pad is used for giving Left, Right, Up and Down information (see Figure.6).

Although I designed this current prototype for using with PCs, Tablet PCs or electronic whiteboards, if I change the user interface, it is also applicable for mobile devices such as mobile phones, PDA s and portable game players etc.

Initial User Study and Evaluations

I held an initial user study with current Positional Prediction text entry prototype in order to know users' typing speed and to get their comments. I used PC mouse, stylus pen with Tablet PC, Logitech dual thumbstick game pad and Nintendo Wii remote controller with three Myanmar native users. All of the participants are familiar with PC but don't have an experience of using stylus pen, dual thumbstick and Wii remote controller. The experiments procedures are 1) explaining the concept of Positional Prediction text entry 2) making demonstration of text input with prototype 3) allowing 15 minutes practice time for each user to learn text input with prototype 4) recording users' typing speeds of short Myanmar message (6 sentences) for 5 times (including error correction time). The Myanmar text used for user study can be seen in Figure.7.

Recorded users' typing speeds are calculated to CPM (Characters per Minute) for evaluation process. The user study results show that the average typing speed of Positional Prediction text entry method to finish six Myanmar sentences by mouse is 3 minutes 15 seconds (32.62 CPM), by stylus pen is 3 minutes 21 seconds

(31.64 CPM), by dual thumbstick game pad is 7 minutes 5 seconds (14.96 CPM) and by Wii remote controller is 5 minutes 30 seconds (19.27 CPM) respectively. With the current prototype, the fastest user's typing speed for each device is: 2 minutes 33 seconds (41.57 CPM) by mouse, 2 minutes 32 seconds (41.84 CPM) by stylus pen, 7 minutes 11 seconds (14.76 CPM) by dual thumbstick and 5 minutes 10 seconds (20.52 CPM) by Wii remote controller. From these results, it is found that the faster typing speed is available with mouse or stylus pen. The first reason is that all of the users are unfamiliar with dual thumbstick game pad and Wii remote. And the second reason is that users' can not directly point or highlight the consonants like in mouse and stylus pen. The third reason is because of the program bugs (e.g. in my current prototype, the mouse pointer movement time with Wii remote controller is slow and sometimes difficult to control).

သူငယ်ချင်း၊
မတွေ့ရတာကြာပြီနော်။
နေကောင်းရဲ့လား။
ငါ့ဖုန်းနံပါတ်အသစ်က၅၀၀၇၄၅၉၊
အားတဲ့အခါဖုန်းပြန်ဆက်ကွာ။
ဒါပဲနော်။

Figure 7. Six Myanmar sentences for user study

To make comparison with the existing Myanmar text input method, I used "Win Myanmar Visual Keyboard" [1]. Actually, "Win Myanmar Visual Keyboard" uses more keys or buttons than my prototype, because it uses QWERTY keyboard interface. The result of my previous user study indicates that the average typing speed of two users for "Win Myanmar Visual Keyboard" is 2 minutes 30 seconds (42.4 CPM) (see Figure. 8). The average typing speed of Positional Prediction text entry method with current proposal (see Figure.9) is 30 % slower than "Win Myanmar Visual Keyboard". But fastest typing speed 41.57 CPM is almost the same as average typing speed of "Win Myanmar Visual Keyboard" (i.e. 42.4 CPM).

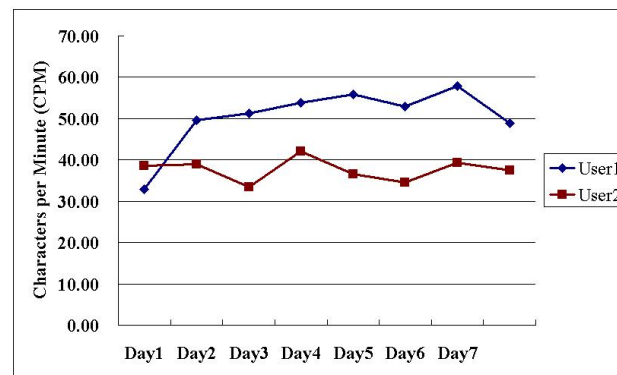


Figure 8. Average typing speed of two Users to finish six Myanmar sentences with "Win Myanmar Visual Keyboard" (experiment results for seven continuous days)

Conclusion

In this paper, I have presented a new predicting text entry method that is applicable for many kinds of mobile devices. I will make further refinements on the

current prototype and follow up an analysis on such as error rate comparison. I also plan to apply this concept for Khmer and Bangla languages in the near future.

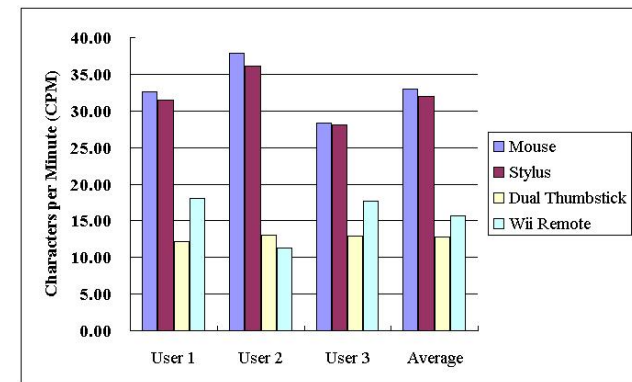


Figure 9. Average typing speed of three Myanmar users with our Positional Prediction text entry prototype

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